

LAB EXPERIMENT RECORD

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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SET 10 – SURVEY RESPONSES ANALYSIS

Aim

To analyze and visualize survey responses using grouped bar charts, stacked bar charts, tables, and an interactive dashboard in RStudio.

Dataset: Monthly Sales

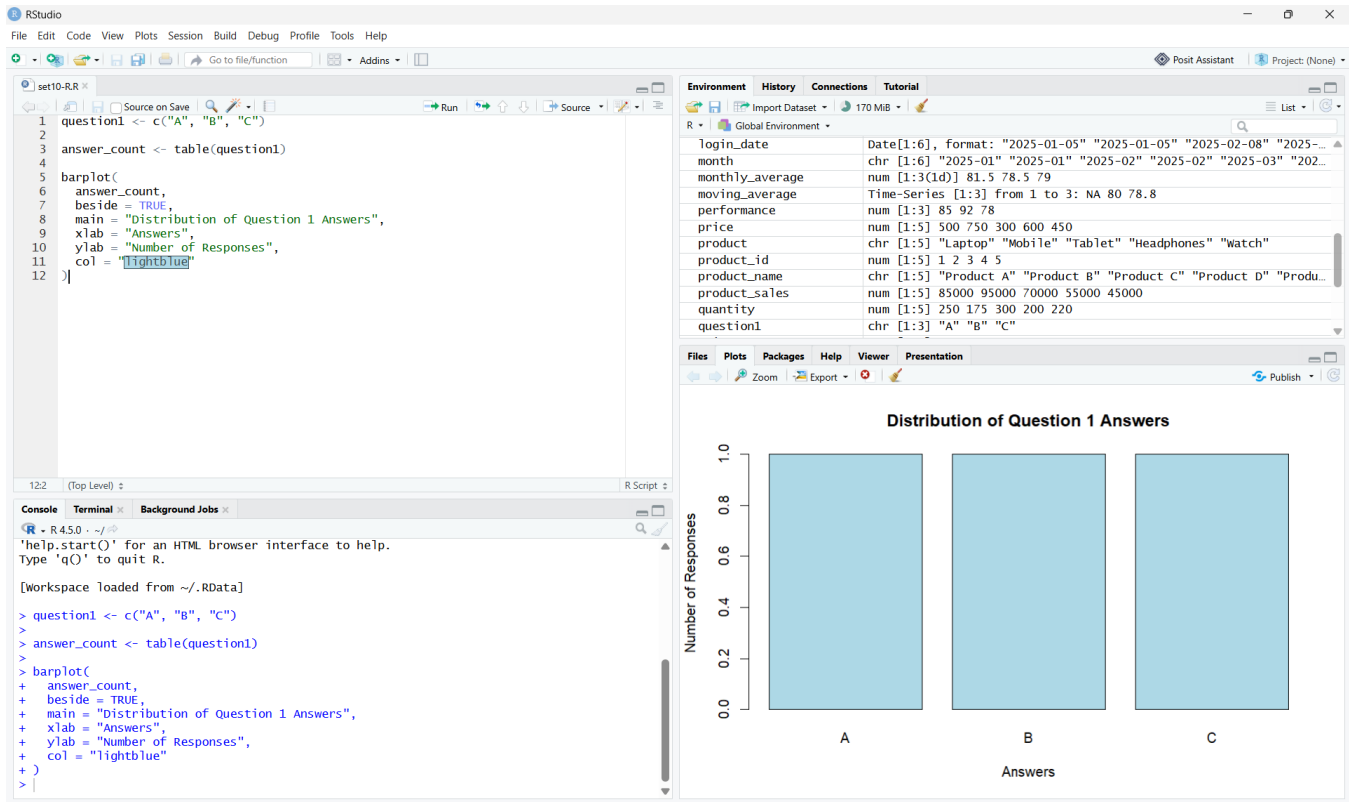
Survey ID Question 1 Question 2 Question 3

1	A	B	C
2	B	A	D
3	C	A	B

PROGRAM 1: Grouped Bar Chart

To create a grouped bar chart to visualize the distribution of answers for Question 1

```
# question1 <- c("A", "B", "C")
answer_count <- table(question1)
barplot(
  answer_count,
  beside = TRUE,
  main = "Distribution of Question 1 Answers",
  xlab = "Answers",
  ylab = "Number of Responses",
  col = "lightblue")
```



Output 1: Monthly Sales Line Chart

Observation: Question 1 has three responses: A, B, and C. Each answer occurs once, so all three categories have equal frequency.

```
question1 <- c("A", "B", "C")
question2 <- c("B", "A", "A")
question3 <- c("C", "D", "B")

response_table <- rbind(
  Q1 = table(factor(question1, levels = c("A", "B", "C", "D"))),
  Q2 = table(factor(question2, levels = c("A", "B", "C", "D"))),
  Q3 = table(factor(question3, levels = c("A", "B", "C", "D")))
)

barplot(
  response_table,
  beside = FALSE,
  main = "Overall Survey Response Distribution",
  xlab = "Questions",
  ylab = "Number of Responses",
  col = c("lightblue", "lightgreen", "orange", "pink"),
  legend.text = c("A", "B", "C", "D")
)
```



Output 2: Monthly Sales Bar Chart

Observation: The stacked bar chart shows the distribution of answers A, B, C, and D across the three questions. Question 2 has two responses of A..

PROGRAM 3: Survey Response Table

To create a table displaying the responses for each survey..

```
# survey_id <- c(1, 2, 3)
question1 <- c("A", "B", "C")
question2 <- c("B", "A", "A")
question3 <- c("C", "D", "B")
survey_table <- data.frame(
  Survey_ID = survey_id,
  Question_1 = question1,
  Question_2 = question2,
  Question_3 = question3
)
survey_table
```

The screenshot shows the RStudio interface. The script editor contains the following R code:

```

22 )
23
24 barplot(
25   response_table,
26   beside = FALSE,
27   main = "Overall Survey Response Distribution",
28   xlab = "Questions",
29   ylab = "Number of Responses",
30   col = c("lightblue", "lightgreen", "orange", "pink"),
31   legend.text = c("A", "B", "C", "D")
32 )
33
34 survey_id <- c(1, 2, 3)
35
36 question1 <- c("A", "B", "C")
37 question2 <- c("B", "A", "A")
38 question3 <- c("C", "D", "B")
39
40 survey_table <- data.frame(
41   Survey_ID = survey_id,
42   Question_1 = question1,
43   Question_2 = question2,
44   Question_3 = question3
45 )
46
47 survey_table

```

The console shows the execution of the code, resulting in the following data frame:

```

>
> question1 <- c("A", "B", "C")
> question2 <- c("B", "A", "A")
> question3 <- c("C", "D", "B")
>
> survey_table <- data.frame(
+   Survey_ID = survey_id,
+   Question_1 = question1,
+   Question_2 = question2,
+   Question_3 = question3
+ )
>
> survey_table
  Survey_ID Question_1 Question_2 Question_3
1         1         A         B         C
2         2         B         A         D
3         3         C         A         B

```

Output 3: Advertising Budget vs Monthly Sales Scatter Plot

Observation: The table clearly displays the responses given by each survey participant for all three questions.

PROGRAM 4: Interactive Dashboard

To develop an interactive dashboard combining the grouped bar chart, stacked bar chart, and survey response table.

```
library(shiny))

output$mytable <- renderTable({
  if (input$view == "Survey Response Table") {
    data.frame(
      Survey_ID = survey_id,
      Question_1 = question1,
      Question_2 = question2,
      Question_3 = question3
    )
  }
})

shinyApp(
  ui = ui,
  server = server
)
```

The screenshot shows the RStudio environment with the Shiny app code in the editor. The code defines a table that is rendered when the 'Survey Response Table' view is selected. The resulting dashboard, titled 'Survey Responses Dashboard', features a 'Select View' dropdown menu. Below the dropdown, a table displays the survey data:

Survey_ID	Question_1	Question_2	Question_3
1.00	A	B	C
2.00	B	A	D
3.00	C	A	B

In the background, a grouped bar chart is visible, showing the distribution of data across categories A, B, C, and D. The chart has a legend on the right side with colored squares corresponding to each category: A (light blue), B (green), C (orange), and D (red).

Output 4: Interactive Monthly Sales Dashboard

Observation: The dashboard allows the user to switch between Question 1 distribution, overall survey responses, and the complete survey response table.

Procedure

- Create vectors for the three survey questions.
- Use `table()` and `barplot()` to visualize Question 1 responses.
- Use `rbind()`, `table()`, and `barplot()` to create the stacked bar chart for all questions.
- Use `data.frame()` to create the survey response table.
- Use Shiny to combine the charts and table into an interactive dashboard.
- Save each question as a separate R script and run them in RStudio.

Result

The survey response data was successfully analyzed using grouped and stacked bar charts and a data table. An interactive dashboard was also successfully developed in RStudio using Shiny.